

Introduction & Motivation

- Marathon performance depends on age, gender, and pacing
- Goal: Predict marathon finish time using machine learning**

Project Objectives

- Compare Linear Regression vs Neural Network
- Evaluate using MAE, RMSE, R^2 , MAPE

Results

Model	MAE (min)	RMSE (min)
Linear Regression	6.4066	9.1491
Neural Network	6.3635	9.1313

Figure 1. Comparison of Linear Regression and Neural Network performance using MAE and RMSE.

Observations

- Neural network slightly outperforms linear regression
- Both models perform similarly

Feature Relationship

- Half-marathon split time shows a strong linear relationship with finish time
- Half-marathon split is the dominant predictor

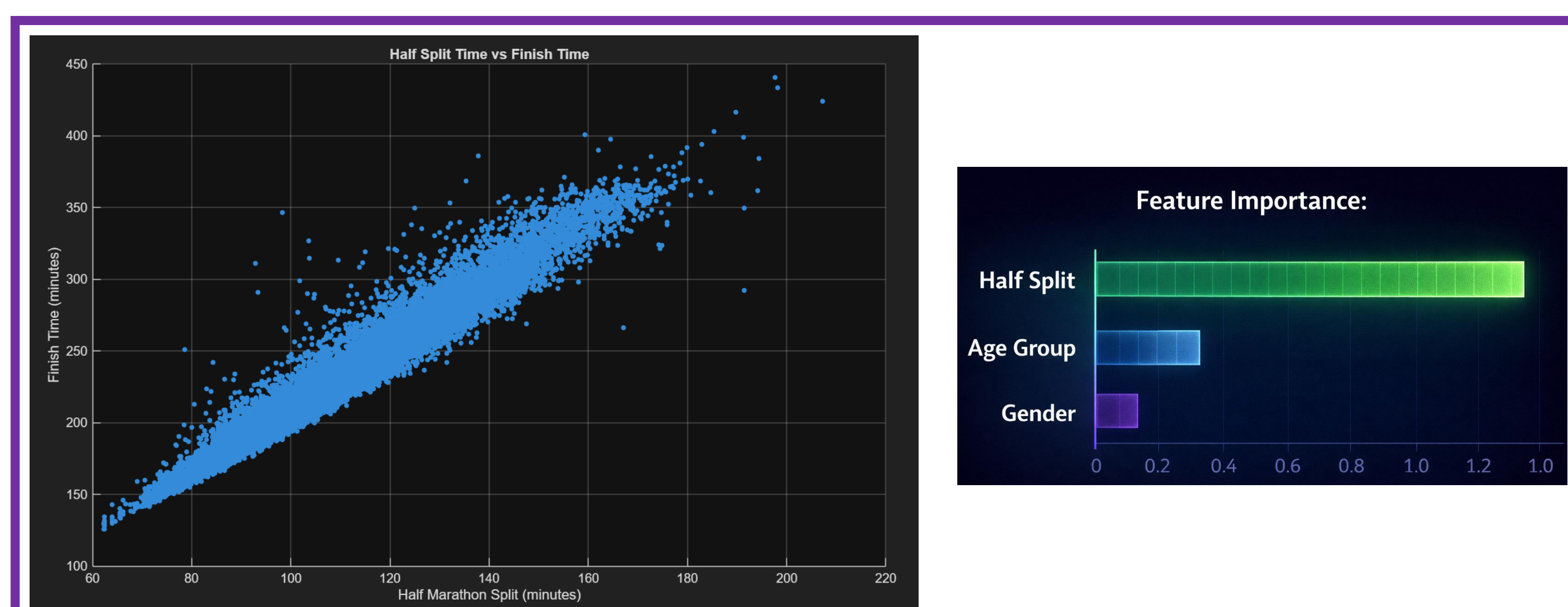


Figure 2. Half-marathon split time vs final finish time showing strong linear correlation.

Methodology

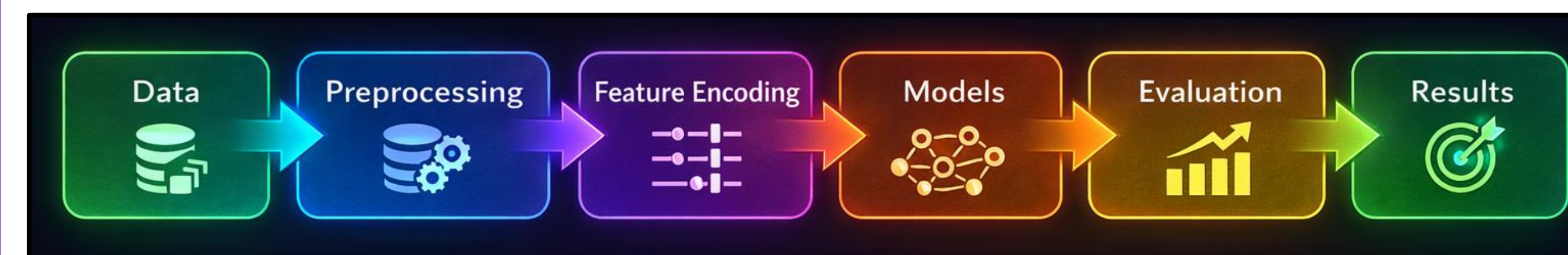


Figure 3. Machine learning pipeline from data processing to final results.

Dataset

- 2023 Boston Marathon → 26,598 runners
- Key Features: age_group, gender, half_time_sec
- Target: finish_net_sec

Models

- Linear Regression** → baseline model
- Neural Network (MLP)** → captures nonlinear patterns

Model Prediction Visualization

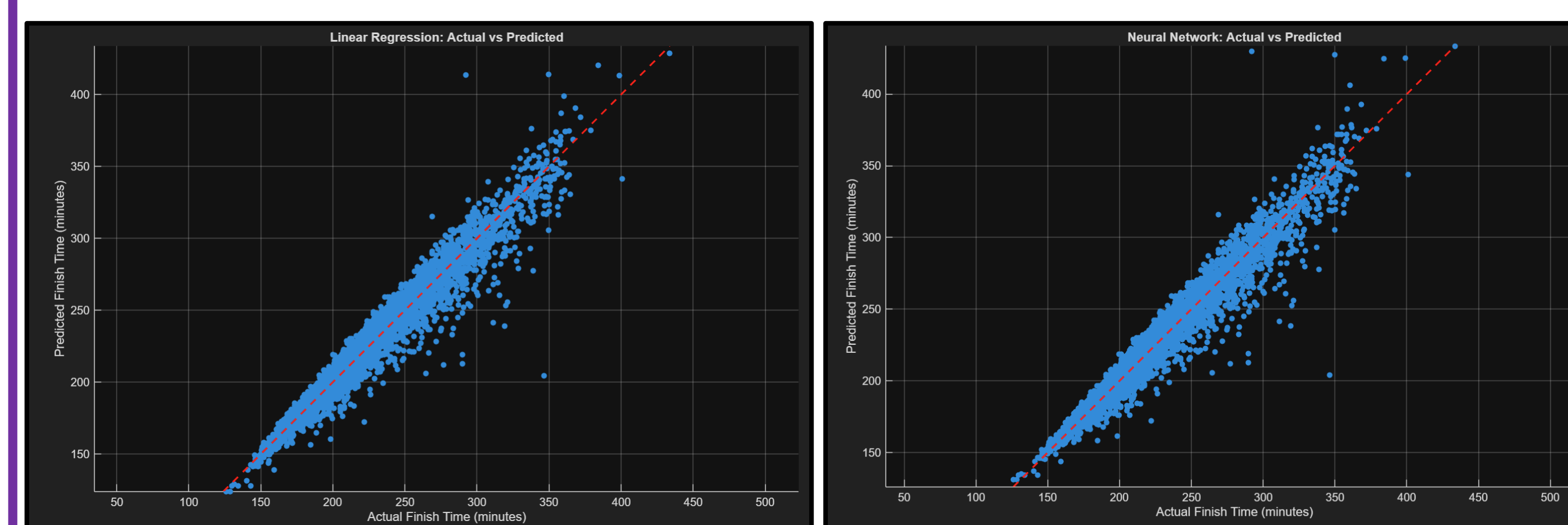


Figure 4. Actual vs. predicted marathon finish times for both models. Points close to the diagonal indicate accurate predictions.

- Both models show strong predictive performance accuracy
- Neural network slightly improves accuracy
- Points close to diagonal → accurate predictions

Acknowledgments

- SCORE Sports Data Repository – [2023 Boston Marathon runners](#). Provides the full dataset of 26,598 runners, including age group, gender, placements, and half/full marathon times, used for all calculations, graphs, and statistical analysis in the project.
- MATLAB (MathWorks) for data analysis and model development.

Decision-Based Extension

- Convert prediction → Over/Under decision
- Thresholds: 180, 240, 300 minutes
- Accuracy: ~94–97%
- Simulates how models can evaluate and calibrate decision thresholds

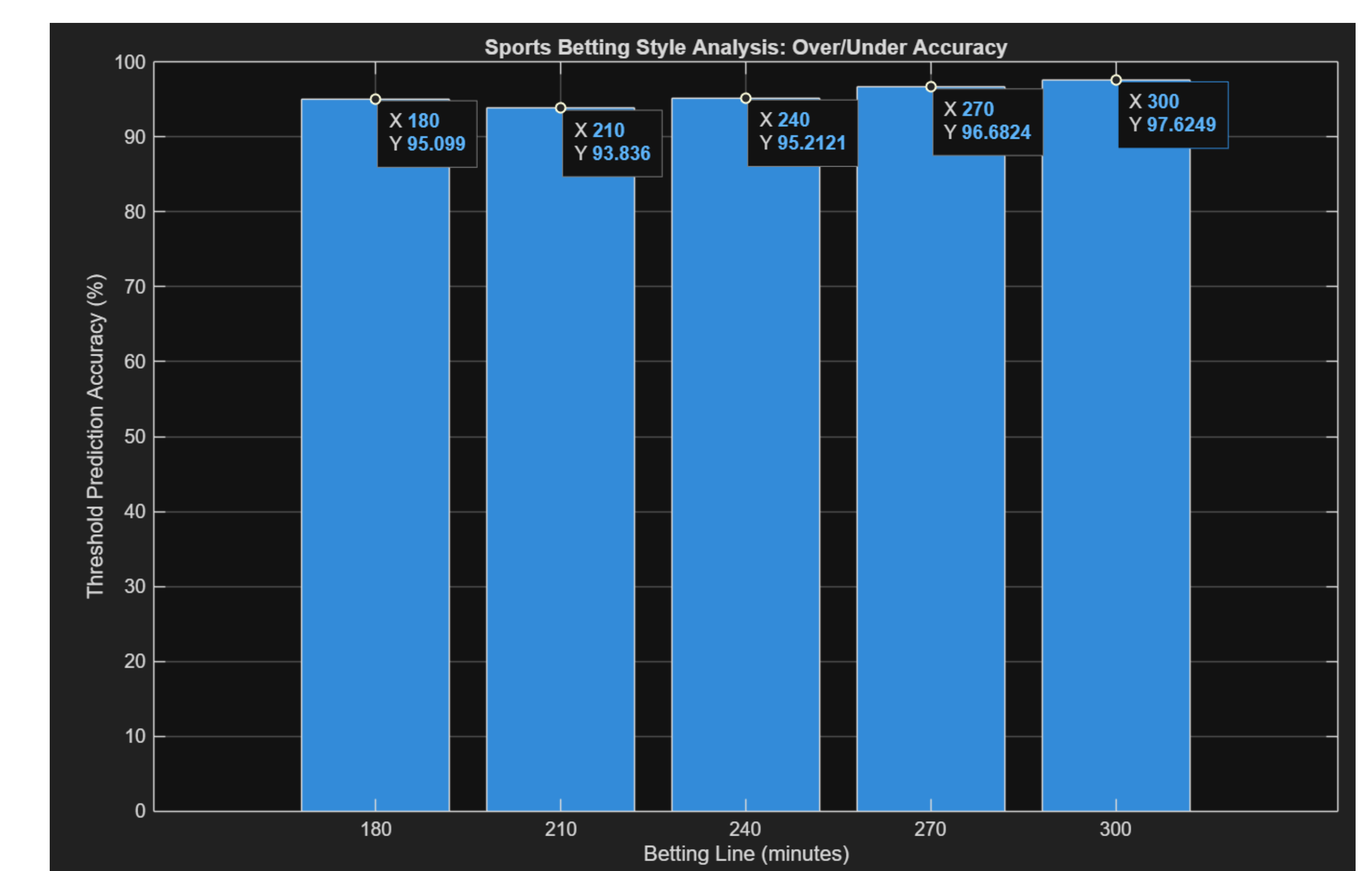


Figure 5. Over/under prediction accuracy across different time thresholds.

- This framework converts regression predictions into actionable decision outcomes.

Discussion

Key Takeaways

- Half-marathon split time is the strongest predictor
- Both models achieve high accuracy
- Neural network provides slight improvement
- Strong linear relationship enables accurate predictions
- Neural networks capture additional nonlinear patterns

Future Work

- Hyperparameter tuning
- Cross-validation
- Explore additional ML models